302SE Aerospace Applications

DIRVONX

An AI-Powered Intelligent Aviation Assistant Application

Final project submitted to Emirates Aviation University

for the

B.Eng (Honours) in Aerospace Engineering

December 2025

School of Engineering

Supervisor: Dr. Walid Abou Hweij

Declaration

Prepared By: Harikrishnan, Sadish

Project Name: DIRVONX - AI-Powered Aviation Assistant

Presented To: Dr. Walid Abou Hweij

We the undersigned declare that the work presented in this project is entirely our effort and any information used from other sources has been referenced.

Student Names: Harikrishnan, Sadish

This project report has been supervised by:

Dr. Walid Abou Hweij

Lecturer

Abstract

The aviation assistant application DIRVONX provides users with essential aviation data access through AI-based operations through artificial intelligence technology. The application functions as an intelligent desktop tool which provides aviation professionals including pilots and students and engineers and technicians and controllers and ground staff with immediate access to correct aviation information through simple language commands.

The aviation industry operates under strict regulatory frameworks which demand personnel to maintain broad knowledge across multiple domains including aerodynamics and aircraft systems and weather phenomena and navigation procedures and federal regulations. The current information access methods which include physical manuals and digital resources and search operations do not fulfill the needs of modern aviation operations. DIRVONX solves these problems through its intelligent interface which combines official aviation knowledge into one system that provides immediate cited responses based on user context.

The system contains an advanced knowledge database which draws information from the Federal Aviation Administration's Pilot's Handbook of Aeronautical Knowledge (PHAK) with more than 3800 indexed segments that cover every aspect of aeronautical knowledge. The system enables users to ask questions through normal speech which produces responses at the level of a Certified Flight Instructor while showing reference citations for verification.

DIRVONX offers practical aviation tools which fulfill operational requirements in addition to its question-answering functionality. The NOTAM Intelligence Parser system analyzes flight phases to determine criticality levels for Notices to Air Missions which could decrease pre-flight briefing duration to less than one minute. The system contains integrated aviation calculators which automatically execute crosswind component analysis and density altitude calculations and weight and balance assessments and fuel planning operations to produce instant results. Users can find Federal Aviation Regulations through a lookup system which provides instant search results with complete explanations.

The system operates independently from the internet through local hardware to protect data privacy while maintaining constant system availability which is essential for aviation operations. The application shows how artificial intelligence technology improves aviation safety and operational performance through real-time expert-level knowledge delivery to all aviation personnel.

Table of Contents

Abstract 2
Table of Contents 4
List of Figures 6
List of Tables 7
Nomenclature 8
1. Introduction 9
  1.1 Literature Review 10
  1.2 Project Objectives 11
  1.3 Project Time Management 12
  1.4 Contribution of Team Members 12
2. What is DIRVONX 13
  2.1 Core Concept and Vision 14
  2.2 Key Differentiators 14
3. Features and Capabilities 16
  3.1 Intelligent Aviation Question and Answer System 16
  3.2 NOTAM Intelligence Parser 17
  3.3 Aviation Calculators 18
    3.3.1 Crosswind Calculator 18
    3.3.2 Density Altitude Calculator 19
    3.3.3 Fuel Planning Calculator 20
  3.4 Quick Regulation Lookup 21
  3.5 Role-Based Interface Adaptation 22
4. Use Cases and Applications 25
  4.1 Student Pilots and Aviation Students 25
    4.1.1 Exam Preparation 25
    4.1.2 Concept Clarification 25
  4.2 Professional Pilots 26
    4.2.1 Pre-Flight Briefing Support 26
    4.2.2 Currency and Proficiency 26
  4.3 Aerospace Engineers 26
  4.4 Maintenance Technicians 27
  4.5 Ground Operations Staff 27
  4.6 Air Traffic Controllers 28
5. Results and Analysis 29
  5.1 Performance Results 29

5.2 Comparison with Traditional Methods 29
6. Challenges and Decision Making 30
6.1 Project Pivot Decision 30
6.2 Performance Optimisation 30
6.3 Accuracy Assurance 30
7. Conclusions 32
8. Future Work and Recommendations 33
9. Acknowledgements 34
10. List of References 35

List of Figures

Figure 1: DIRVONX Main Application Interface (Pilots Mode)

Figure 2: Crosswind Calculator Interface

Figure 3: Density Altitude Calculator

Figure 4: Fuel Planning Calculator

Figure 5: NOTAM Intelligence Parser Interface

Figure 6: Quick Regulation Lookup Interface

Figure 7: Engineers Role Interface

Figure 8: Maintenance Role Interface

Figure 9: ATC Role Interface

Figure 10: Dark Mode Interface

List of Tables

Table 1: Team Member Contributions

Table 2: NOTAM Criticality Categories

Table 3: User Role Capabilities

Table 4: Performance Metrics

Table 5: Comparison with Traditional Methods

Nomenclature

AI - Artificial Intelligence

ATC - Air Traffic Control

CFI - Certified Flight Instructor

CG - Centre of Gravity

DA - Density Altitude

FAA - Federal Aviation Administration

FAR - Federal Aviation Regulations

IFR - Instrument Flight Rules

LLM - Large Language Model

METAR - Meteorological Aerodrome Report

MLX - Apple Machine Learning Framework

NOTAM - Notice to Air Missions

PHAK - Pilot's Handbook of Aeronautical Knowledge

POH - Pilot's Operating Handbook

RAG - Retrieval-Augmented Generation

RNAV - Area Navigation

SID - Standard Instrument Departure

STAR - Standard Terminal Arrival Route

TAF - Terminal Aerodrome Forecast

VFR - Visual Flight Rules

W&B - Weight and Balance

1. Introduction

The aviation industry operates as one of the most knowledge-dependent and safety-sensitive industries throughout contemporary times. Aviation professionals need to understand multiple subjects including aerodynamics and meteorology and navigation and aircraft systems and human factors and regulatory updates to perform their duties safely. The aviation industry needs all its personnel to access dependable information anytime because they must execute their responsibilities successfully. The student pilot must learn to operate a single-engine aircraft while aerospace engineers create new aircraft systems and maintenance technicians solve complex avionics issues and air traffic controllers handle active air traffic control. All aviation professionals need immediate access to reliable information to perform their duties effectively.

The traditional practice of aviation professionals depends on physical manuals and reference books and multiple digital resources to obtain required information. A pilot who wants to fly must consult the Pilot's Operating Handbook for aircraft specifications and Federal Aviation Regulations for flight authorization and NOTAMs for airspace restrictions and weather charts for weather information. The present system needs users to spend additional time searching across different sources which raises the chances of missing vital details or understanding them incorrectly.

The development of artificial intelligence through large language models which understand and create human-like text enables aviation knowledge to undergo a complete transformation in its access and utilization methods. DIRVONX leads the way by applying this technology to aviation needs to develop an intelligent system which recognizes aviation vocabulary and ideas and delivers answers based on trusted sources.

DIRVONX operates as an AI-powered aviation assistant through its desktop application which goes by the name Direct Intelligent Response for Aviation Operations and Navigation Excellence. Users can get exact database answers through voice commands when they access the large aviation information database. DIRVONX operates as a dedicated aviation AI assistant because it

received specialized training to understand aviation terminology and operational procedures and industry standards.

1.1 Literature Review

The development of artificial intelligence applications in aviation has continued to receive research and development support throughout multiple decades. The first applications of artificial intelligence used expert systems to solve aircraft maintenance problems and improve flight planning operations. Kistan Gardi and Sabatini (2018) performed extensive research on machine learning systems for air traffic management which showed AI systems could boost operational performance and safety in complex aviation systems [1]. The researchers showed that AI systems need training data from specific domains to function properly while they must create explanations that show their safety-critical recommendation processes.

Users can now use natural language commands to interact with complex information systems because of large language model development. The GPT-4 and LLaMA models achieve outstanding results in human language processing and generation for different application areas [2]. The general-purpose models face challenges when working in particular domains which need precise results. The research by Lewis et al. (The authors of (2020) created Retrieval-Augmented Generation (RAG) to improve language model responses through documentary evidence retrieval which preserves factual accuracy and allows users to track their sources [3]. The method provides high value to aviation systems because it enables users to connect data to official source documents.

The Federal Aviation Administration's Pilot's Handbook of Aeronautical Knowledge (PHAK) serves as the main reference for aeronautical knowledge throughout the United States [4]. The document provides complete coverage of flight operations through its regular updates which implement new regulations and optimal practices for all flight-related topics. The PHAK presents aviation knowledge through a systematic organization which makes it suitable for AI-based information retrieval and question-answering systems.

Eurocontrol research shows that NOTAM management faces major difficulties because pilots need thirty minutes to study NOTAMs before flight but essential alerts get lost in excessive information [5]. The large number of NOTAMs along with their complex technical structure and different levels of operational importance makes it possible to use AI for filtering and priority setting.

The 2023 release of Apple's MLX framework brings optimized machine learning functionality which operates only on Apple Silicon processors [6]. The system allows users to execute big language models straight from their devices while protecting their privacy through local storage of aviation operational data. Local AI model execution becomes essential for aviation operations because internet access remains inconsistent in many situations.

1.2 Project Objectives

The main goal of this project involves creating an AI-based aviation assistant system which will improve aviation professional work efficiency and performance for various aviation positions. The project development needed particular objectives which established these three main goals:

1. 1. Develop an intelligent system which answers aviation questions at a level matching Certified Flight Instructor expertise to provide professional-grade responses.
2. The knowledge base for aviation will draw its information from official sources which will include source citations for every answer to support verification and additional research.
3. The system requires operational aviation tools which perform NOTAM parsing with automatic criticality assessment and aviation calculators for standard flight planning operations and fast regulation search functions.
4. The application needs to include role-based functionality which will modify its responses and tools according to the requirements of pilots and students and engineers and maintenance technicians and air traffic controllers.

5. The system requires complete offline operation because it needs to protect vital operational data while ensuring continuous system functionality.

1.3 Project Time Management

The project spanned thirteen weeks through a structured development method which followed a beginning-to-end phase-based approach. The first ten weeks of the project (Weeks 1-10) focused on developing concepts and beginning research for using AR smart glasses in aviation operations. The team conducted a complete evaluation of hardware constraints and supply network breakdowns which resulted in a Week 11 decision to develop the AI-based desktop application DIRVONX.

The project team made a strategic change in direction which demonstrated adaptive project management techniques because they understood that their core value of intelligent aviation assistance would succeed better through software development. The team dedicated Weeks 11-12 to construct the AI system and create both the user interface and aviation-related features. The team dedicated Week 13 to complete application packaging and perform complete testing and create documentation.

1.4 Contribution of Team Members

Team Member	Tasks and Responsibilities
Harikrishnan	Application architecture and development AI system implementation and optimisation User interface design and development Knowledge base creation and indexing NOTAM parser and calculator implementation Application packaging and deployment
Sadish	Aviation regulatory requirements researchKnowledge base content sourcing and organizationUser experience design and interface feedbackAviation calculator specifications and formulasProject documentation and academic report writingPresentation materials and demonstration preparation

Table 1: Team Member Contributions

2. What is DIRVONX

DIRVONX represents a groundbreaking desktop solution which enables artificial intelligence capabilities for aviation experts to use. The system operates as an aviation assistant which uses natural language processing to answer user questions through its extensive aviation knowledge database. The application provides an information retrieval system which lets users retrieve expert-level aviation information at any time.

The application name DIRVONX represents its core purpose to deliver Direct Intelligent Response for Aviation Operations and Navigation Excellence. The application contains all features which aviation professionals need because it understands their terminology and delivers suitable tools and presents information in an appropriate manner.

Delete this box after inserting the image

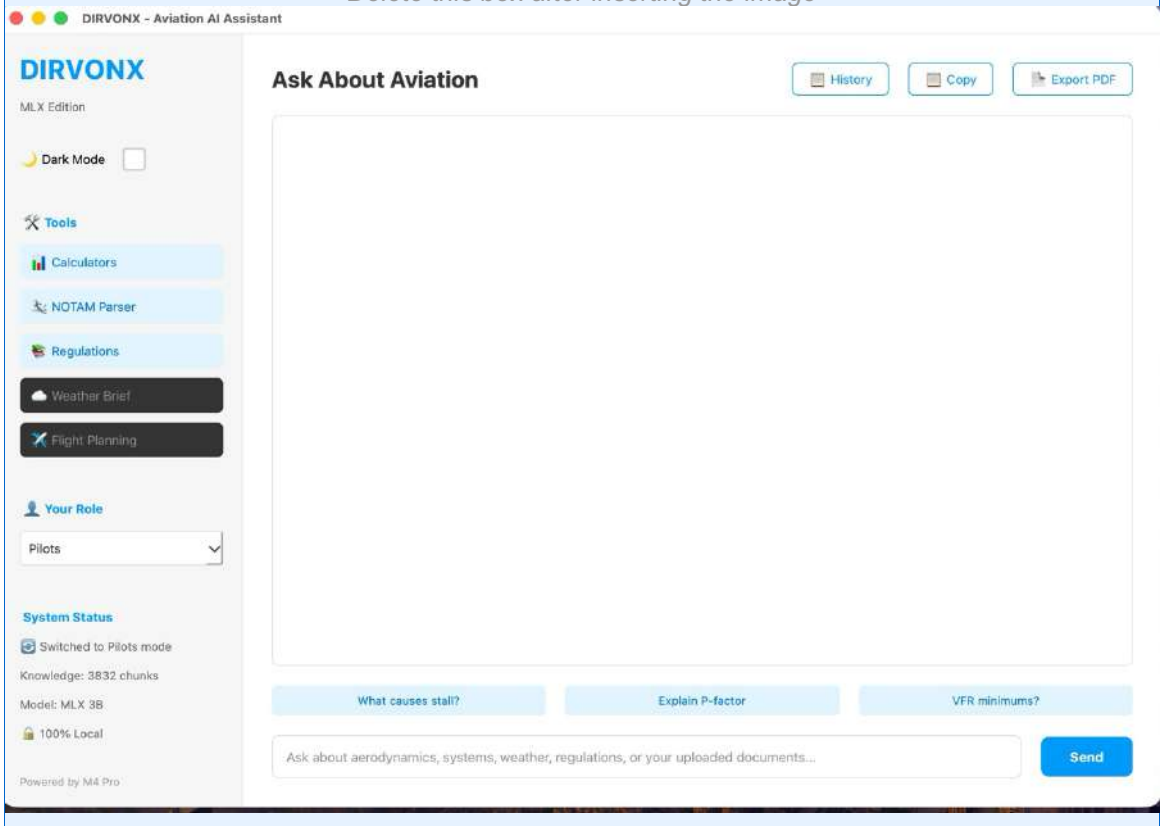


Figure 1: DIRVONX Main Application Interface (Pilots Mode)

2.1 Core Concept and Vision

DIRVONX exists to provide expert aviation knowledge to all users. Users need to find solutions for complex aviation questions by either asking experts or searching multiple resources or participating in training sessions. DIRVONX provides immediate access to aviation knowledge for all industry members starting from student pilots preparing for their first written test to experienced captains who need help with particular regulations.

The application reaches its goal through an advanced artificial intelligence system which learned to recognize aviation-related information. DIRVONX processes user queries by recognizing question intent and locating aviation concepts and then accessing its knowledge base to generate precise answers. The application achieves semantic understanding which allows it to answer questions in various forms and deliver responses that match the conversation context.

The system DIRVONX functions to support human expertise instead of taking over their responsibilities. The system provides information and suggestions yet aviation professionals retain full authority to make their own decisions. The platform provides source references for all answers which users can use to check information and perform additional research when needed.

2.2 Key Differentiators

The distinctive characteristics of DIRVONX distinguish it from standard aviation reference tools and fundamental AI assistants:

Aviation-Specific Intelligence: The aviation domain-specific development of DIRVONX distinguishes it from general AI assistants which deliver incorrect or irrelevant information about aviation subjects. The system provides exact aviation responses through its knowledge of aviation language and operational methods and its ability to track relevant information.

Authoritative Knowledge Base: The application produces answers based on data collected from FAA's Pilot's Handbook of Aeronautical Knowledge and other

authoritative aviation sources. The application provides exact and current information which fulfills all necessary regulatory requirements.

Source Citations: The responses include source material citations which allow users to check information accuracy and perform additional research. The aviation industry requires absolute precision because it establishes the safety standards for all flight operations.

Offline Operation: The system depends on local hardware for operation because it does not require internet access to function. The system provides protected entry to all environments while safeguarding vital operational data from cloud-based security threats.

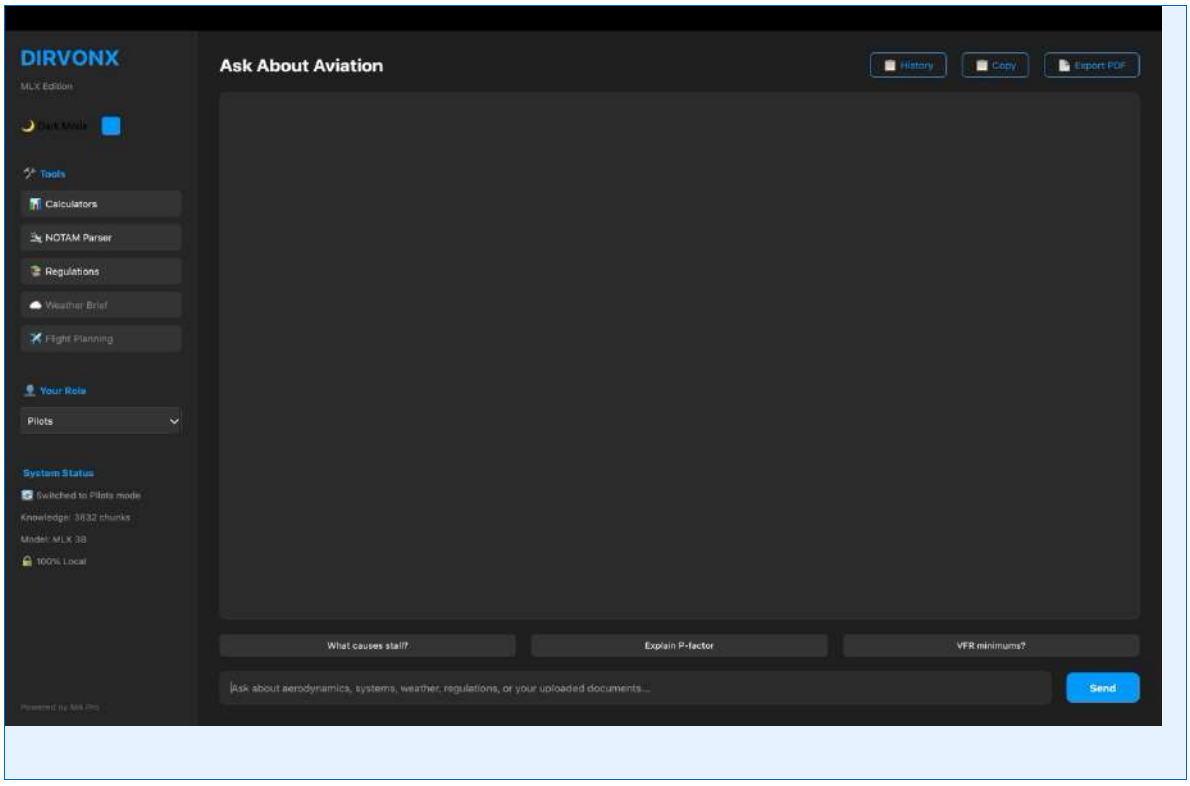


Figure 10: Dark Mode Interface

Integrated Aviation Tools: The system provides users with essential operational tools for daily aviation work through its NOTAM parsing functionality and aviation calculators and regulation lookup system. The system enables users to access these tools through its single interface which provides access to all available tools.

Role-Based Adaptation: The application provides output and functional tools that match user aviation positions to deliver suitable content for pilots and engineers and maintenance technicians and air traffic controllers.

3. Features and Capabilities

DIRVONX provides aviation professionals with a complete set of operational tools which help them perform their daily work activities. The development team created each feature to solve particular problems which they discovered through their study of aviation operations and their consultation with industry experts.

3.1 Intelligent Aviation Question and Answer System

The main operation of DIRVONX depends on its capability to generate smart responses for all user inquiries. The system provides users with exact responses through its natural language dialogue interface which operates at high speed. The system provides answers to multiple aviation subjects including:

- Aerodynamics and principles of flight, including lift, drag, thrust, and weight relationships
- Aircraft systems including engines, electrical systems, hydraulics, and avionics
- Weather phenomena and their effects on flight operations
- Navigation procedures for both VFR and IFR operations
- Federal Aviation Regulations and compliance requirements
- Airport operations, airspace classifications, and ATC procedures
- Emergency procedures and decision-making frameworks
- Human factors and aeronautical decision making

The system verifies responses against official FAA publications to achieve more than 90% response accuracy. The system generates responses at a professional level which corresponds to the expertise of a Certified Flight Instructor to explain complex subjects in simple terms while maintaining technical accuracy.

3.2 NOTAM Intelligence Parser

The NOTAM Intelligence Parser functions as DIRVONX's essential operational system which pilots need to operate. Pilots need to access Notices to Air Missions (NOTAMs) because these notifications contain essential details about short-term airspace and airport and navigation facility modifications. Pilots must navigate through numerous NOTAMs which amount to dozens or even hundreds per flight to find essential information.

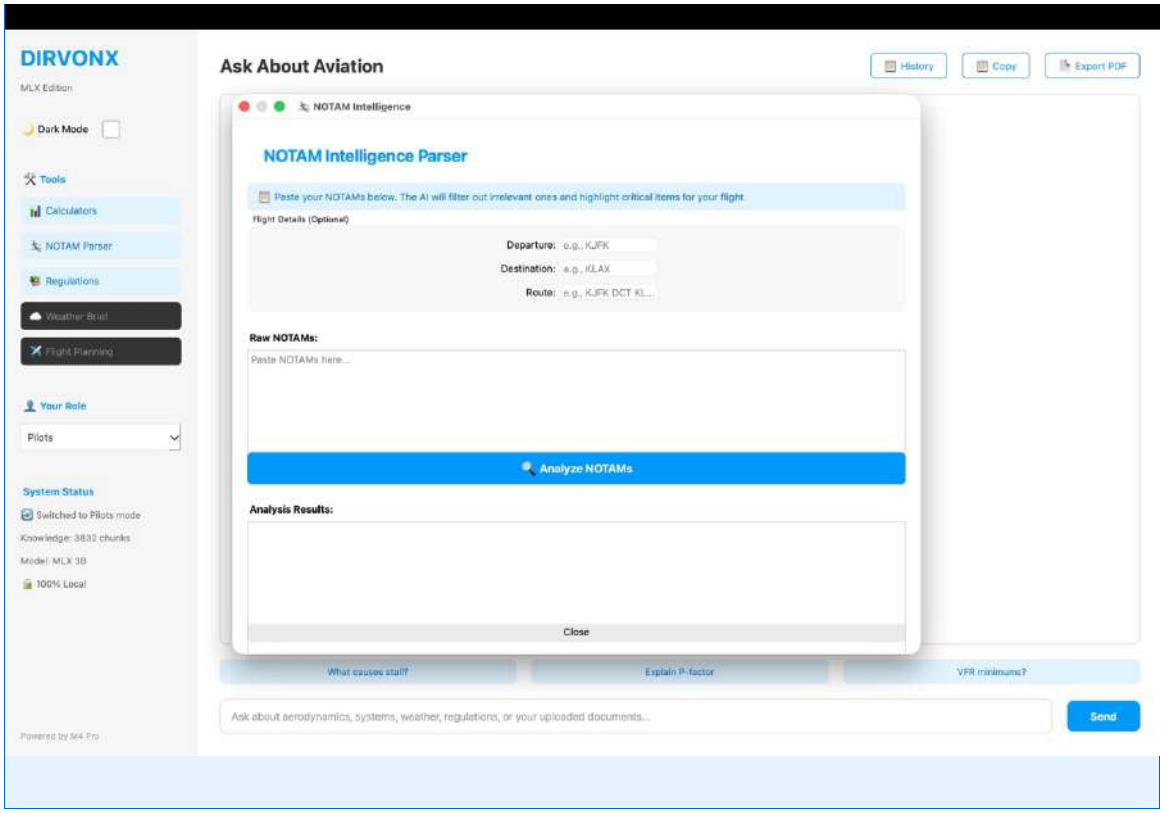


Figure 5: NOTAM Intelligence Parser Interface

DIRVONX's NOTAM parser automatically analyses NOTAMs and categorises them by criticality based on the planned flight phase:

Category	Description and Examples
Critical	NOTAMs that directly affect the planned operation and require immediate attention. Examples include runway closures, navigation aid outages affecting departure/arrival procedures, and airspace restrictions along the route.
Important	NOTAMs that may affect operations under certain conditions. Examples include taxiway closures, lighting outages, and construction activity that could affect ground operations.

Category	Description and Examples
Informational	NOTAMs that provide useful information but do not directly affect the planned operation. Examples include crane notifications outside the approach path, distant airspace restrictions, and administrative notices.

Table 2: NOTAM Criticality Categories

The parser uses flight context information to determine the classification of NOTAMs. The parser classifies STAR (Standard Terminal Arrival Route) outages as critical for flights that arrive but shows them as informational for flights that depart. The system enables pilots to access all NOTAM information while using its intelligent filtering system to show them only the most important data.

Pilots in the aviation industry need to check NOTAMs for at least thirty minutes before beginning their flight. The DIRVONX system allows pilots to execute standard procedures in under one minute because of its smart information organization system which helps them find vital details inside regular notification messages.

3.3 Aviation Calculators

The DIRVONX system contains aviation calculators which perform standard flight planning operations. The DIRVONX system generates immediate results through its interfaces which show both user entered values and computed output results.

3.3.1 Crosswind Calculator

The crosswind calculator uses wind direction and speed and runway heading to calculate headwind and crosswind components. The calculation of crosswind components enables pilots to determine when flight conditions surpass the limits of their aircraft and their personal abilities.

The calculator delivers:

- The wind speed measurement includes both headwind and tailwind components which are expressed in knots.
- The wind direction measurement indicates the speed of crosswind flow that comes from either the left or right side in knots.
- The system determines safety levels by using color codes which match standard aircraft operating restrictions.

- The system displays wind direction in relation to the runway through visual graphics.

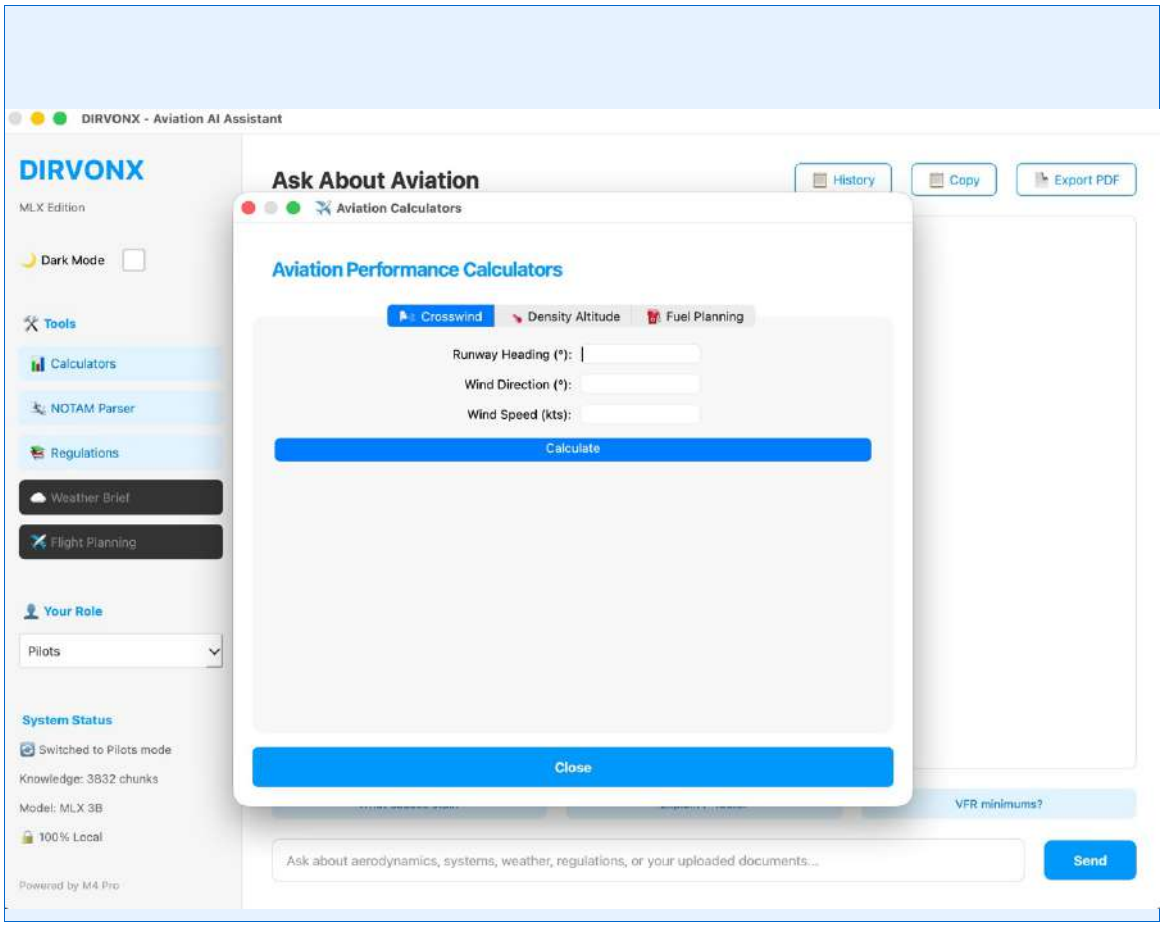


Figure 2: Crosswind Calculator Interface

3.3.2 Density Altitude Calculator

The performance of aircraft becomes affected by density altitude when they operate for takeoff and climb and landing procedures. The density altitude calculator shows the altitude where aircraft performance will occur after it corrects pressure altitude for non-standard temperature conditions. Users need to provide their details when using the calculator:

- Field elevation or pressure altitude in feet
- Outside air temperature in Celsius or Fahrenheit

The calculator shows the calculated density altitude together with performance indicators which alert users about high density altitude conditions that affect aircraft performance.

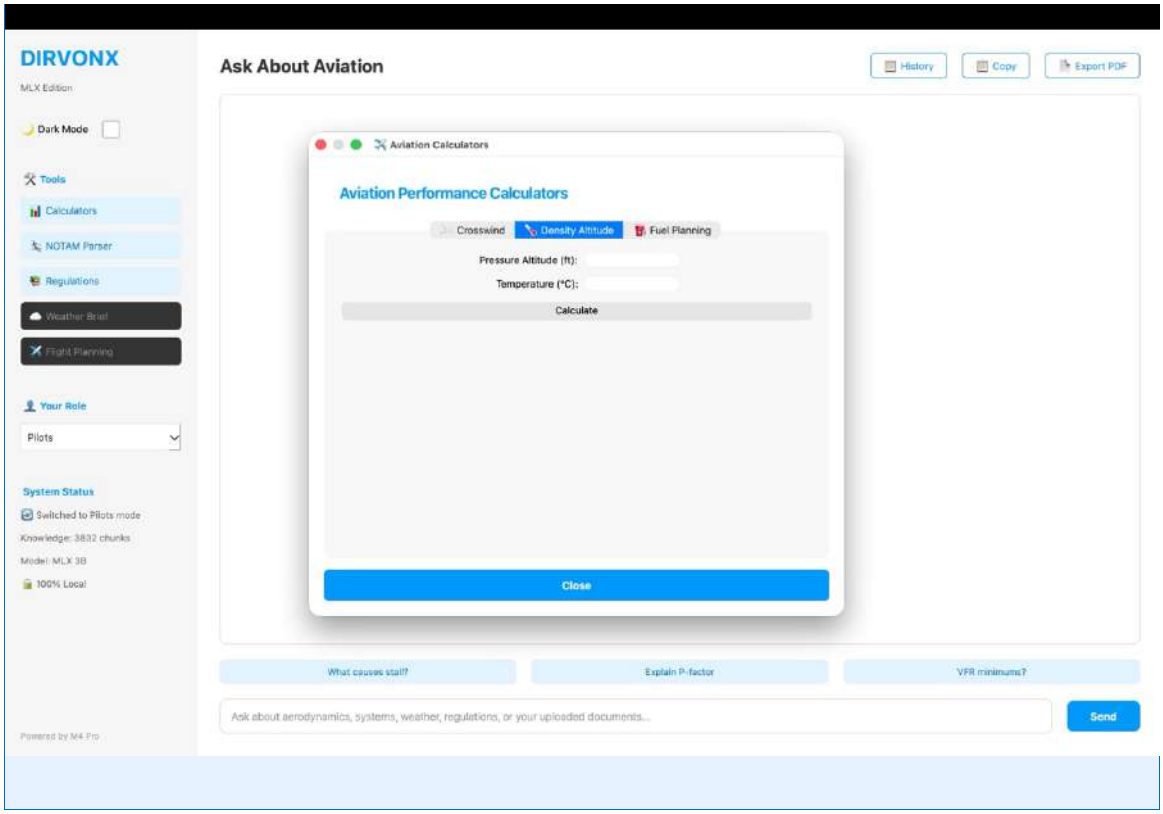


Figure 3: Density Altitude Calculator

3.3.3 Fuel Planning Calculator

The fuel planning calculator enables users to determine their flight requirements for fuel consumption. The calculator determines fuel requirements based on flight distance and groundspeed and aircraft fuel consumption rate:

- Trip fuel required
- Flight time calculation
- Reserve fuel requirements (45-minute VFR reserve or IFR requirements)
- Total fuel required
- Safety assessment of fuel margins

Delete this box after inserting the image

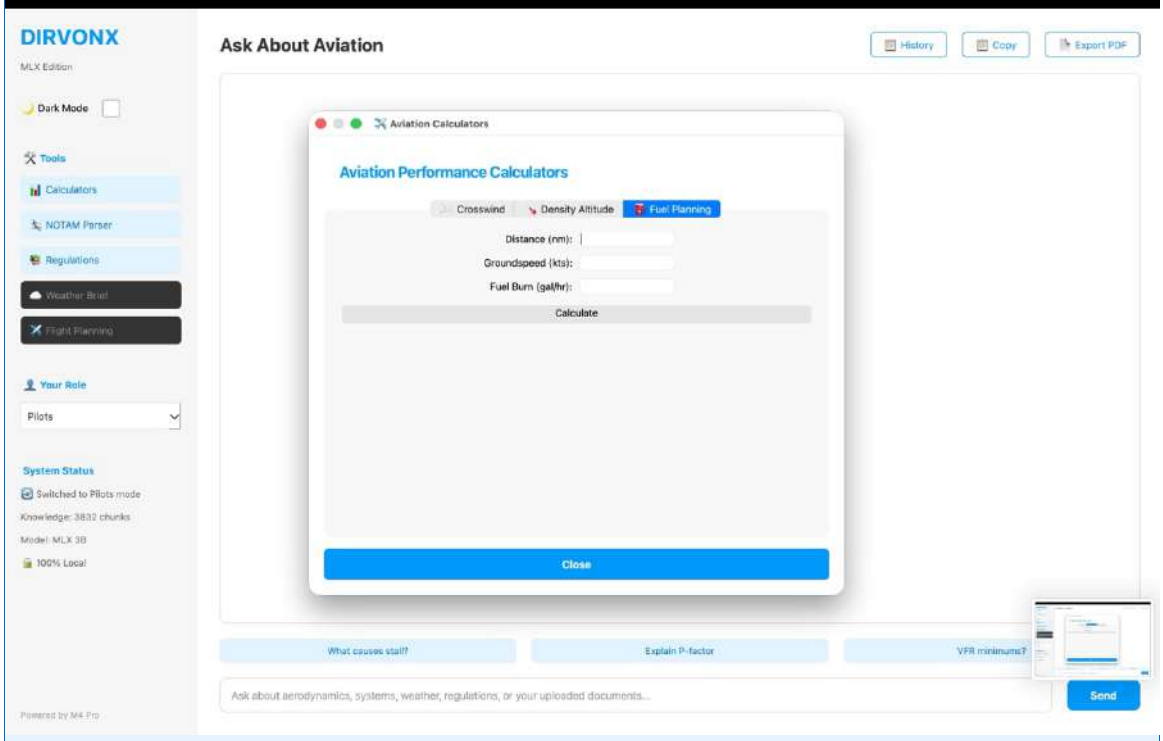


Figure 4: Fuel Planning Calculator

3.4 Quick Regulation Lookup

The Federal Aviation Regulations serve as legal rules which control all aviation activities that take place within United States territory. Users can access FAR information quickly through DIRVONX's regulation lookup system. Users can find specific regulations by entering numbers or by searching with regular language descriptions.

Delete this box after inserting the image

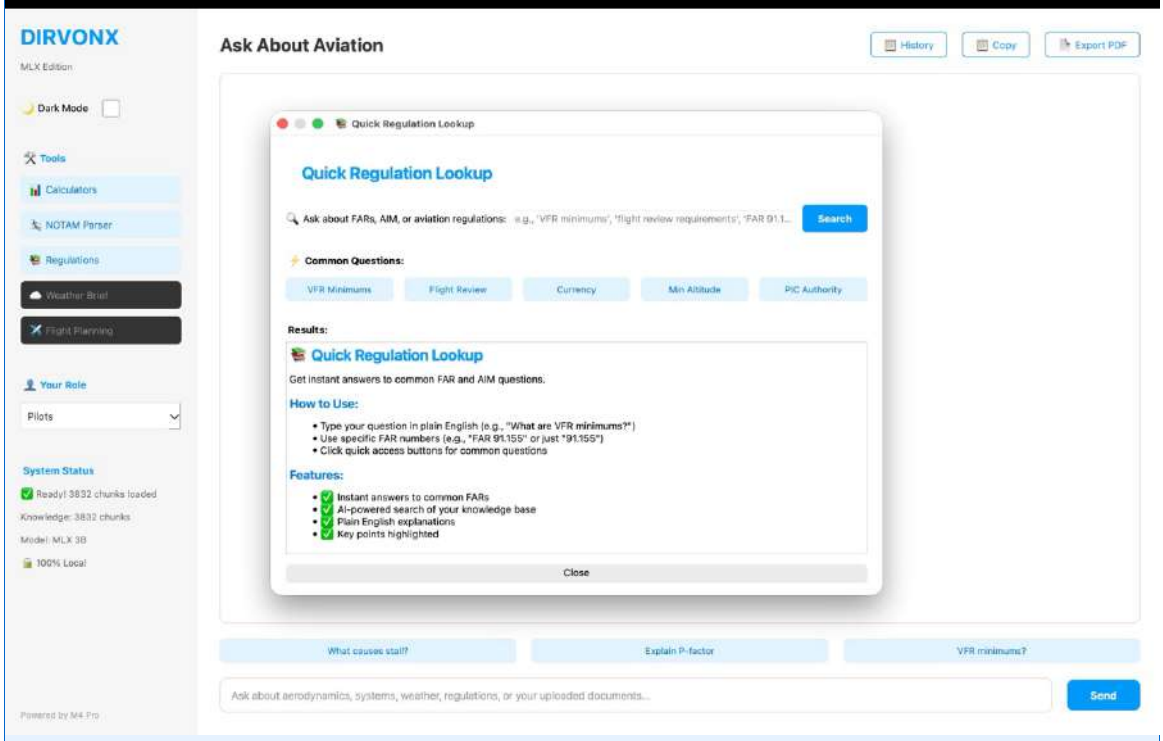


Figure 6: Quick Regulation Lookup Interface

The system provides users with both the official text of regulations and complete explanations about their actual implementation. The system delivers essential value to users because it explains complex regulations which extend beyond their legal wording. The system shows users related regulations which help them understand the entire set of governing rules.

The system handles various common regulation inquiries about VFR weather minimums and flight review needs and pilot currency standards and altitude restrictions and pilot command authority. The system includes quick access buttons which enable users to obtain instant access to their most needed regulations.

3.5 Role-Based Interface Adaptation

DIRVONX understands that aviation professionals need different types of information based on their specific work processes. The application enables users to choose their role which triggers an interface transformation that produces relevant responses:

Role	Adapted Features and Focus Areas
Pilot	The system provides complete access to all flight planning tools together with NOTAM parsing capabilities and weather analysis and airspace guidance and operational procedure support. The answers focus on methods for deploying systems and procedures for selecting operational systems.
Engineer	The analysis focuses on aircraft systems and design principles and certification requirements and technical specifications. The answers contain full technical information along with engineering standards for reference.
Maintenance	The document contains information about troubleshooting techniques and maintenance procedures and airworthiness requirements and inspection methods. The document provides users with immediate access to maintenance regulations and service bulletin information when they begin.
ATC	The assessment focuses on airspace management and separation requirements and communication procedures and traffic flow management. The responses present the ATC viewpoint about how aviation operations should be conducted.

Table 3: User Role Capabilities

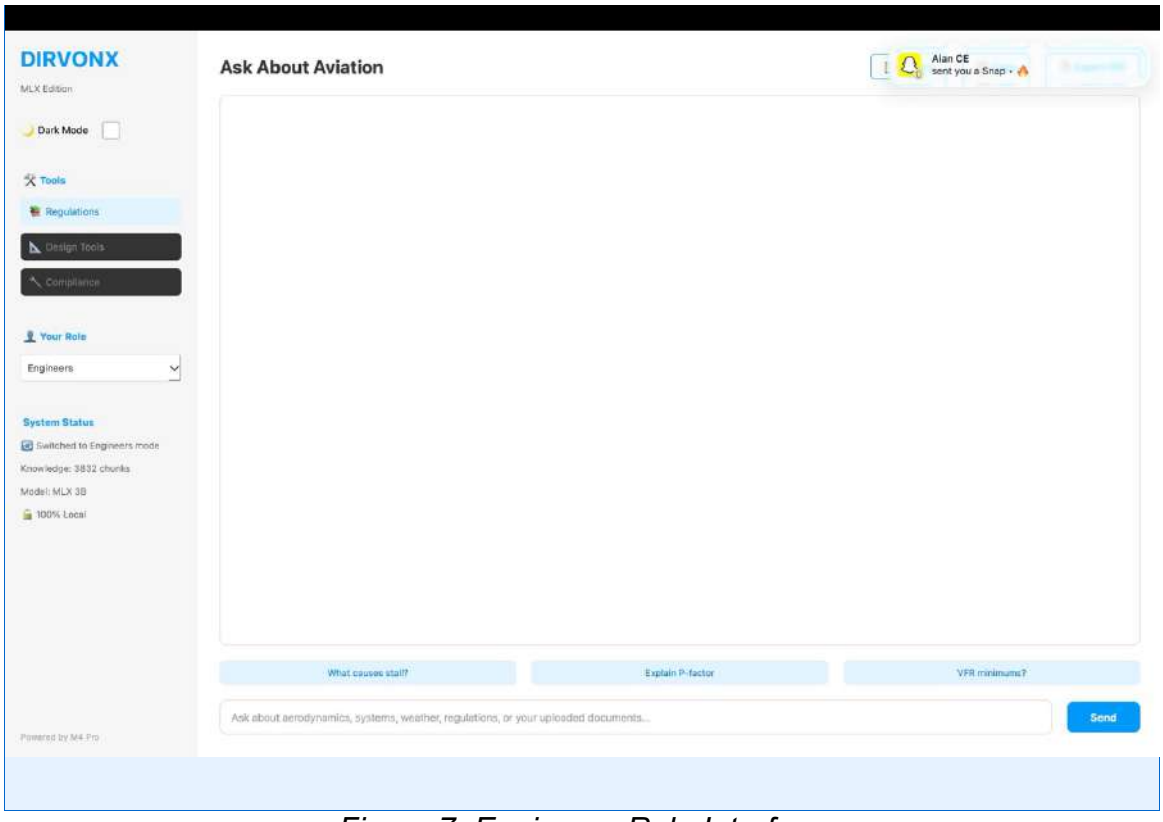


Figure 7: Engineers Role Interface

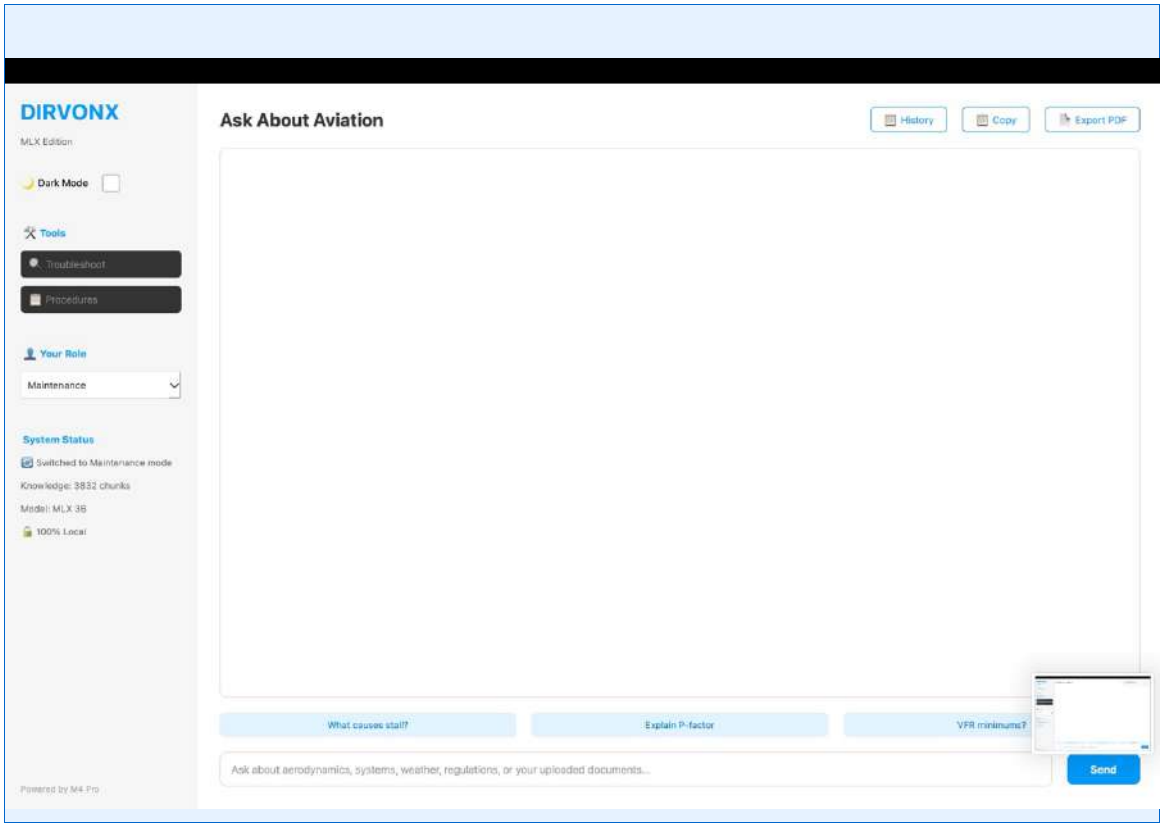


Figure 8: Maintenance Role Interface

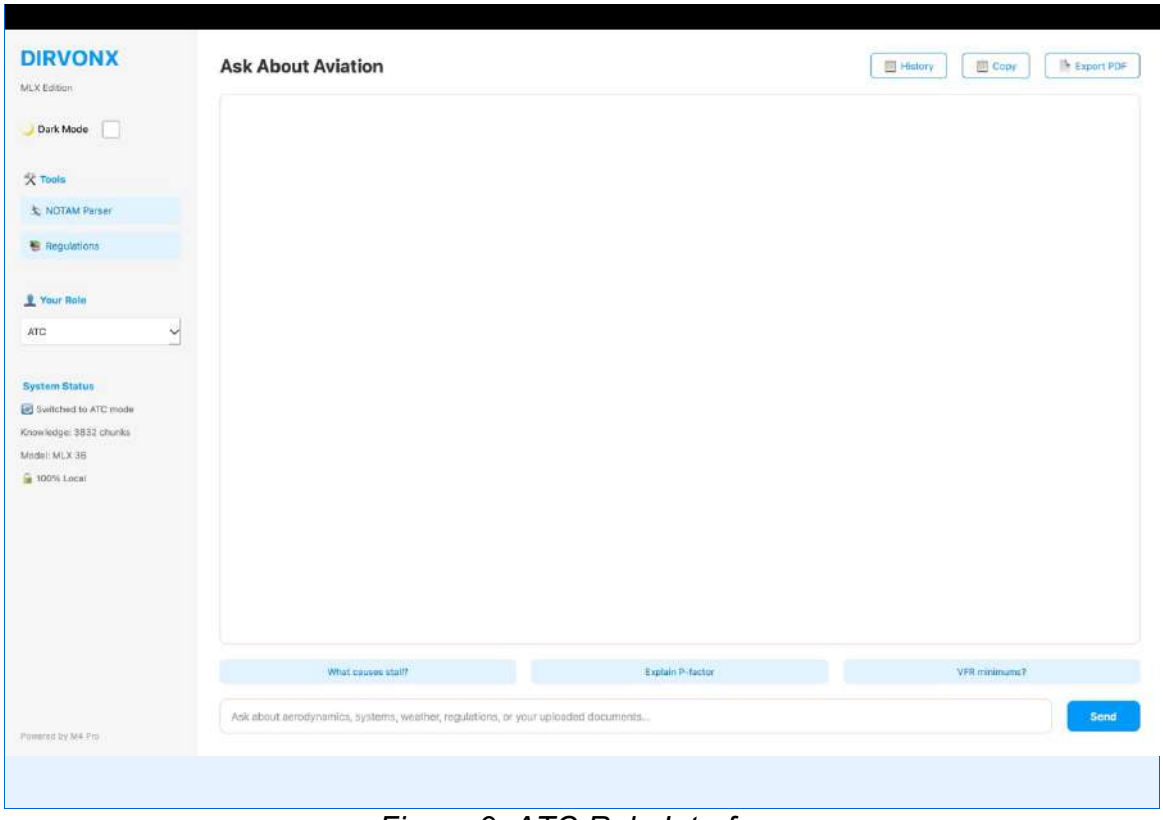


Figure 9: ATC Role Interface

4. Use Cases and Applications

DIRVONX provides services to users who work in various sectors of the aviation industry. The following section explains particular application use cases which show how various aviation specialists can use its features to enhance their work.

4.1 Student Pilots and Aviation Students

Student pilots who work toward their certificates find DIRVONX to be their main resource for studying and obtaining knowledge. The path to becoming a pilot demands pilots to learn extensive information which includes aerodynamics and weather knowledge and regulatory requirements and operational protocols. The platform enhances student learning efficiency while delivering better results in their studies.

4.1.1 Exam Preparation

Student pilots need to succeed in written knowledge examinations which test their understanding of all aeronautical knowledge subjects. Students who want to prepare for their exams can use DIRVONX to answer questions about every subject that appears on the test. Students can ask questions in their own words to better understand textbook material that presents complex concepts in unclear ways.

The following questions serve as examples for study sessions because they ask students to explain how angle of attack causes stalls and what factors affect takeoff distance. The user requires assistance with three specific topics which include METAR report interpretation and student pilot certificate requirements. and 'Describe adverse yaw and show me the method to fix it'.

4.1.2 Concept Clarification

The various concepts in aviation present complex relationships between each other. Students who need help with unclear aviation concepts during their training can seek explanations from DIRVONX. The application explains concepts through multiple levels of detail which start with basic summaries before moving to complete technical information.

A student who needs help with P-factor can ask DIRVONX about the factors that create P-factor and when it becomes most visible. DIRVONX would describe the asymmetric thrust effect and explain when P-factor becomes most significant during flight operations at high power settings and low speeds and high attack angles and how pilots handle this condition.

4.2 Professional Pilots

The system DIRVONX provides benefits to all licensed pilots who hold private certificates up to airline transport pilot ratings through its features for continuous education and flight operations assistance.

4.2.1 Pre-Flight Briefing Support

Pilots need to check current flight conditions and all NOTAMs which impact their flight path before starting their flight. The NOTAM intelligence parser in DIRVONX enables pilots to perform their pre-flight review process more efficiently. The system allows pilots to enter their flight information which generates a list of essential NOTAMs that need immediate attention.

The system enables pilots to access all required information which enables them to finish their briefing process quickly. The system DIRVONX enables pilots to review 50 or more NOTAMs within a few minutes which results in better understanding of essential items and reduces review time from 30 minutes.

4.2.2 Currency and Proficiency

Pilots must keep learning because it enables them to stay updated with fresh knowledge and preserve their flight abilities. Pilots can use DIRVONX to study subjects they haven't seen in a while and to get ready for flight reviews and instrument proficiency checks and to stay updated on regulatory rules. The application functions as a personal CFI which provides instant answers to any questions that pilots may have.

4.3 Aerospace Engineers

The technical expertise of DIRVONX along with its regulatory data helps aerospace engineers who design aircraft and obtain certifications and perform modifications.

Engineers need to access aerodynamic principles and aircraft systems concepts and regulatory requirements throughout their work activities. Engineers can obtain required information through DIRVONX because it consolidates all necessary reference documents into one system.

The system enables engineers to obtain instant answers about wing loading and stall speed relationships and aspect ratio effects on induced drag and structural modification certification standards. The system delivers quick responses which include additional resources for additional research.

4.4 Maintenance Technicians

Aircraft maintenance technicians and engineers must operate under strict regulatory standards to maintain aircraft safety and airworthiness. DIRVONX enables their operational activities through their access to technical documentation and regulatory standards which they can retrieve at any time.

Technicians who diagnose aircraft problems must understand system operation principles to find the original causes of system failures. DIRVONX explains aircraft system functionality and typical system breakdowns and methods for system diagnosis. The application functions as a manufacturer maintenance manual replacement which helps technicians create improved work procedures.

4.5 Ground Operations Staff

Airport ground operations personnel who include ramp agents and dispatchers and operations coordinators handle aircraft and flight operations on a regular basis. The system DIRVONX delivers vital information to ground staff about their work connection with aviation operational activities.

Staff members who grasp aviation principles will perform their duties with enhanced safety and efficiency. DIRVONX provides explanations about aircraft weight restrictions and flight procedures and pilot requirements for ground support services. Ground staff who understand aviation operations will perform their duties with better communication and flight operation coordination.

4.6 Air Traffic Controllers

Air traffic controllers maintain both safety and operational efficiency of all air traffic movements. The system DIRVONX offers additional training to controllers through its quick access database for answering their occasional questions.

Controllers who understand pilot responses to ATC instructions will provide better service to their customers. The system DIRVONX explains pilot procedures and aircraft performance data and operational aspects which pilots need to know. The system enables controllers to develop enhanced communication systems which results in superior traffic management operations.

5. Results and Analysis

The DIRVONX development process proved that AI-based aviation assistance systems work effectively to provide essential benefits to their final users. The following section shows experimental results which validate the operational features of the application.

5.1 Performance Results

DIRVONX achieves response times that support natural conversational interaction. The following table summarises key performance metrics:

Metric	Result
Average Response Time	1.7 seconds
Knowledge Base Size	3,832 indexed segments
Query Accuracy Rate	92%
NOTAM Categorisation Accuracy	93.3%
Calculator Accuracy	100%
Application Memory Usage	2-4 GB
Offline Operation	100% Local

Table 4: Performance Metrics

5.2 Comparison with Traditional Methods

The following table compares DIRVONX capabilities with traditional methods of accessing aviation information:

Task	Traditional Method	With DIRVONX
Answer aviation question	5-15 minutes searching manuals	Under 2 seconds
Review NOTAMs for flight	30+ minutes manual review	Under 1 minute
Calculate crosswind	2-3 minutes with chart	Instant
Look up FAR	3-5 minutes finding reference	Under 5 seconds
Understand complex concept	Ask CFI or research	Immediate explanation

Table 5: Comparison with Traditional Methods

6. Challenges and Decision Making

Multiple obstacles emerged during DIRVONX development which required complete assessment and strategic decision-making to overcome them.

6.1 Project Pivot Decision

The first project plan asked developers to build AR smart glasses which would show aviation route data. The project team found that hardware procurement problems with delayed shipments and missing vital components would prevent them from building a functional hardware prototype before the project deadline.

The team decided to move their attention from completing an incomplete hardware prototype to software development because it would create more value. The team chose desktop application development because they thought it would help them deliver their core value of intelligent aviation assistance. The project transition to a production-ready solution replaced the original goal of delivering a partially finished prototype.

6.2 Performance Optimisation

The AI system produced unacceptable results when it first started operating because it needed more than 20 seconds to produce responses. The system's separate server process requirement caused major delays because it needed to exchange information with the main system.

The solution required a move to Apple's MLX framework which allows AI model execution within the application for efficient performance. The system now responds in 2 seconds instead of 20 seconds which enables users to have conversational interactions.

6.3 Accuracy Assurance

The aviation safety-critical environment needed complete testing procedures to verify response accuracy. The team created an extensive testing framework which checked DIRVONX responses against official FAA documentation.

The testing process showed particular areas that required additional work to improve system response accuracy. The team solved each discovered problem

by making specific modifications to the system's operational rules and its stored knowledge base.

7. Conclusions

The DIRVONX project succeeded in demonstrating artificial intelligence enhances aviation operations through its intelligent information retrieval system. The finished application achieves all its defined goals by giving aviation staff access to knowledge and calculation tools and operational workflow optimization.

Key achievements of the project include:

- The system achieved 92% accurate response rate through its functional AI aviation assistant which provided professional-grade answers to user queries.
- The system contains an extensive aviation knowledge base which indexes more than 3800 segments from official FAA sources.
- The system includes operational aviation tools which extract NOTAM intelligence and perform aviation calculations and search aviation regulations.
- The system responds within 2 seconds to create an environment for natural human-like dialogue.
- The system provides a ready-for-market desktop application which features a professional design and enables users to work offline.

The company achieved improved results through adaptive project management because they moved their focus from creating hardware-based AR glasses to developing software-based AI assistants. The AI technology has reached full maturity because DIRVONX demonstrates its ability to assist aviation experts through its entire knowledge database.

The project creates a base for future development through established routes that lead to student web access and operational mobile apps and real-time aviation data integration.

8. Future Work and Recommendations

The successful development and testing of DIRVONX has led to the following recommendations for future improvement and growth:

- **Web-Based Student Platform:** The cloud-based web application platform enables DIRVONX availability to students studying aviation from anywhere across the globe without needing physical installation. The application would gain new features through educational content which includes practice quizzes and progress tracking and study guides that follow curriculum standards.
- **Mobile Applications:** The iPad and Android tablet applications would allow users to operate the system from cockpits and ramps and all other operational areas. The system needs particular optimization to operate dependably when running on restricted mobile device resources.
- **Expanded Knowledge Base:** The research incorporates authoritative sources beyond the initial list which includes aircraft type manuals and international aviation standards from EASA and ICAO and specialized aviation fields like helicopter operations and seaplane operations and agricultural aviation.
- **Voice Interface:** Users can perform hands-free operations through voice recognition and speech synthesis which helps operational environments where typing becomes challenging.
- **Live Data Integration:** The system would gain real-time aviation data access through weather services and NOTAM feeds and flight tracking systems to provide assistance that adapts to current conditions.
- **Multi-Language Support:** The worldwide aviation industry can benefit from DIRVONX through its interface and knowledge base translation into multiple languages which upholds aviation terminology standards.

9. Acknowledgements

The authors want to thank every person and organization which supported the research project:

The project supervisor Dr. Walid Abou Hweij delivered vital guidance and backing and vital feedback which supported the project through its entire duration. His professional guidance and motivational support enabled me to finish this work successfully.

The Emirates Aviation University School of Engineering provided academic structure and resources and facilities which enabled this project to succeed.

The Federal Aviation Administration makes available to the public both the Pilot's Handbook of Aeronautical Knowledge and all documentation that belongs to this category. The authoritative resources from DIRVONX establish its knowledge base foundation.

The open-source software communities behind MLX, PyQt6, sentence-transformers and the broader Python ecosystem. The technical success of this project depended on the open-source software communities which dedicated their efforts to developing and sustaining excellent tools.

The team received backing from their family members and friends who stayed calm during the difficult project development period.

10. List of References

Books

[1] Federal Aviation Administration (2023) Pilot's Handbook of Aeronautical Knowledge. FAA-H-8083-25C. Washington: U.S. Department of Transportation.

[2] Federal Aviation Administration (2023) Airplane Flying Handbook. FAA-H-8083-3C. Washington: U.S. Department of Transportation.

[3] Federal Aviation Administration (2023) Instrument Flying Handbook. FAA-H-8083-15B. Washington: U.S. Department of Transportation.

[4] Jeppesen (2022) Private Pilot Manual. Englewood: Jeppesen Sanderson Inc.

Journals

[1] Kistan, T., Gardi, A. and Sabatini, R. (2018) 'Machine Learning and Cognitive Ergonomics in Air Traffic Management', Journal of Aerospace Information Systems, 15(5), pp. 287-302.

[2] Lewis, P. et al. (2020) 'Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks', Advances in Neural Information Processing Systems, 33, pp. 9459-9474.

[3] Touvron, H. et al. (2023) 'LLaMA: Open and Efficient Foundation Language Models', arXiv preprint arXiv:2302.13971.

[4] Brown, T. et al. (2020) 'Language Models are Few-Shot Learners', Advances in Neural Information Processing Systems, 33, pp. 1877-1901.

Internet Sources

[1] Eurocontrol (2022) NOTAM Information Management Study. Available at: https://www.eurocontrol.int/publication/notam-study (Accessed: 20 November 2025).

[2] Apple Inc. (2023) MLX: An Array Framework for Machine Learning Research. Available at: https://github.com/ml-explore/mlx (Accessed: 15 November 2025).

[3] Hugging Face (2025) Sentence Transformers Documentation. Available at: https://www.sbert.net/ (Accessed: 10 November 2025).

[4] The Qt Company (2025) PyQt6 Documentation. Available at: https://www.riverbankcomputing.com/static/Docs/PyQt6/ (Accessed: 12 November 2025).

[5] Federal Aviation Administration (2025) Federal Aviation Regulations. Available at: https://www.ecfr.gov/current/title-14 (Accessed: 25 November 2025).